

MSDM5004

Numerical Methods and Modeling in Science

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Lecture 6

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4. SOR method

Successive Over-Relaxation

$$x_i^{(k)} = (1 - \omega)x_i^{(k-1)} + \frac{\omega}{a_{ii}} \left[- \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^n a_{ij}x_j^{(k-1)} + b_i \right]$$

for certain choices of positive ω

Idea Linear combination of the result of Gauss-Seidel iteration and $x_i^{(k-1)}$

Over-relaxation: $\omega > 1$

Analysis of the SOR method

$$D\mathbf{x}^{(k)} = (1 - \omega)D\mathbf{x}^{(k-1)} + \omega(L\mathbf{x}^{(k)} + U\mathbf{x}^{(k-1)} + \mathbf{b})$$

for certain choices of positive ω

Linear combination of the result of Gauss-Seidel iteration

$$D\mathbf{x}^{(k)} = L\mathbf{x}^{(k)} + U\mathbf{x}^{(k-1)} + \mathbf{b}$$

and

$$D\mathbf{x}^{(k-1)}$$

write $A = D - L - U$

$$D\mathbf{x}^{(k)} = (1 - \omega)D\mathbf{x}^{(k-1)} + \omega(L\mathbf{x}^{(k)} + U\mathbf{x}^{(k-1)} + \mathbf{b})$$

$$(D - \omega L)\mathbf{x}^{(k)} = [(1 - \omega)D + \omega U]\mathbf{x}^{(k-1)} + \omega\mathbf{b}$$

The SOR method

$$\mathbf{x}^{(k)} = (D - \omega L)^{-1}[(1 - \omega)D + \omega U]\mathbf{x}^{(k-1)} + \omega(D - \omega L)^{-1}\mathbf{b}$$

The SOR method can be written in the form

$$\mathbf{x}^{(k)} = T_{\omega} \mathbf{x}^{(k-1)} + \mathbf{c}_{\omega}$$

with

$$T_{\omega} = (D - \omega L)^{-1} [(1 - \omega)D + \omega U]$$

$$\mathbf{c}_{\omega} = \omega (D - \omega L)^{-1} \mathbf{b}$$

Example

- The linear system $A\mathbf{x} = \mathbf{b}$ given by

$$\begin{aligned}4x_1 + 3x_2 &= 24 \\3x_1 + 4x_2 - x_3 &= 30 \\-x_2 + 4x_3 &= -24\end{aligned}$$

- Compare the iterations from the Gauss-Seidel method and the SOR method with $\omega = 1.25$ using $\mathbf{x}^{(0)} = (1, 1, 1)^t$ for both methods.

For each $k = 1, 2, \dots$, the equations for the Gauss-Seidel method are

$$x_1^{(k)} = -0.75x_2^{(k-1)} + 6$$

$$x_2^{(k)} = -0.75x_1^{(k)} + 0.25x_3^{(k-1)} + 7.5$$

$$x_3^{(k)} = 0.25x_2^{(k)} - 6$$

and the equations for the SOR method with $\omega = 1.25$ are

$$x_1^{(k)} = -0.25x_1^{(k-1)} - 0.9375x_2^{(k-1)} + 7.5$$

$$x_2^{(k)} = -0.9375x_1^{(k)} - 0.25x_2^{(k-1)} + 0.3125x_3^{(k-1)} + 9.375$$

$$x_3^{(k)} = 0.3125x_2^{(k)} - 0.25x_3^{(k-1)} - 7.5$$

Gauss-Seidel Iterations

k	0	1	2	3	...	7
$x_1^{(k)}$	1	5.250000	3.1406250	3.0878906		3.0134110
$x_2^{(k)}$	1	3.812500	3.8828125	3.9267578		3.9888241
$x_3^{(k)}$	1	-5.046875	-5.0292969	-5.0183105		-5.0027940

SOR Iterations ($\omega = 1.25$)

k	0	1	2	3	...	7
$x_1^{(k)}$	1	6.312500	2.6223145	3.1333027		3.0000498
$x_2^{(k)}$	1	3.5195313	3.9585266	4.0102646		4.0002586
$x_3^{(k)}$	1	-6.6501465	-4.6004238	-5.0966863		-5.0003486

Remark: The exact solution is $x=(3,4,-5)^t$.

For the iterates to be accurate to 7 decimal places,

- the Gauss-Seidel method requires 34 iterations,
- as opposed to 14 iterations for the SOR method with $\omega = 1.25$.

5. Convergence of the iterative methods

Theorem

For any $\mathbf{x}^{(0)} \in \mathbb{R}^n$, the sequence $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ defined by

$$\mathbf{x}^{(k)} = T\mathbf{x}^{(k-1)} + \mathbf{c}, \quad \text{for each } k \geq 1,$$

converges to the unique solution of $\mathbf{x} = T\mathbf{x} + \mathbf{c}$ if and only if $\rho(T) < 1$

(similar to fixed-point iteration)

The **spectral radius** $\rho(A)$ of a matrix A is defined by

$$\rho(A) = \max |\lambda|, \quad \text{where } \lambda \text{ is an eigenvalue of } A.$$

(For complex $\lambda = \alpha + \beta i$, we define $|\lambda| = (\alpha^2 + \beta^2)^{1/2}$.)

Diagonally dominant matrices

Definition

The $n \times n$ matrix A is said to be **diagonally dominant** when

$$|a_{ii}| \geq \sum_{\substack{j=1, \\ j \neq i}}^n |a_{ij}| \quad \text{holds for each } i = 1, 2, \dots, n.$$

A diagonally dominant matrix is said to be **strictly diagonally dominant** when the inequality in (6.10) is strict for each n , that is, when

$$|a_{ii}| > \sum_{\substack{j=1, \\ j \neq i}}^n |a_{ij}| \quad \text{holds for each } i = 1, 2, \dots, n.$$

Theorem A strictly diagonally dominant matrix A is nonsingular.

Example of diagonally dominant matrix

$$A = \begin{bmatrix} 7 & 2 & 0 \\ 3 & 5 & -1 \\ 0 & 5 & -6 \end{bmatrix}$$

matrix A is strictly diagonally dominant because

$$|7| > |2| + |0|, \quad |5| > |3| + |-1|, \quad \text{and} \quad |-6| > |0| + |5|.$$

Convergence of the Jacobi and Gauss-Seidel methods

Theorem

If A is strictly diagonally dominant, then for any choice of $\mathbf{x}^{(0)}$, both the Jacobi and Gauss-Seidel methods give sequences $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ that converge to the unique solution of $A\mathbf{x} = \mathbf{b}$. ■

Convergence of the SOR method

Theorem (Kahan)

If $a_{ii} \neq 0$, for each $i = 1, 2, \dots, n$, then $\rho(T_\omega) \geq |\omega - 1|$. This implies that the SOR method can converge only if $0 < \omega < 2$.

Theorem (Ostrowski-Reich)

If A is a positive definite matrix and $0 < \omega < 2$, then the SOR method converges for any choice of initial approximate vector $\mathbf{x}^{(0)}$.

Theorem

If A is positive definite and tridiagonal, then $\rho(T_g) = [\rho(T_j)]^2 < 1$, and the optimal choice of ω for the SOR method is

$$\omega = \frac{2}{1 + \sqrt{1 - [\rho(T_j)]^2}}.$$

With this choice of ω , we have $\rho(T_\omega) = \omega - 1$. 

6. Other iterative methods

Linear system $A\mathbf{x} = \mathbf{b}$

Conjugate gradient (CG) method

Matrix A is symmetric and positive definite

The solution of the linear system is the minimizer of the quadratic function

$$g(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A\mathbf{x} - \mathbf{b}^T \mathbf{x}$$

Large linear systems with symmetric, positive definite matrices are commonly solved using the CG method.

The set of vectors $\{\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(n)}\}$ are A -conjugate if

$$\langle \mathbf{v}^{(i)}, A\mathbf{v}^{(j)} \rangle = 0, \quad \text{if } i \neq j.$$

The CG algorithm

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + t_k \mathbf{v}^{(k)}$$

$$t_k = \frac{\langle \mathbf{v}^{(k)}, \mathbf{b} - A\mathbf{x}^{(k-1)} \rangle}{\langle \mathbf{v}^{(k)}, A\mathbf{v}^{(k)} \rangle}$$

The set of vectors $\{\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(n)}\}$ are generated from the residual $\mathbf{r} = \mathbf{b} - A\mathbf{x}$: $\{\mathbf{r}^{(1)}, \mathbf{r}^{(2)}, \dots, \mathbf{r}^{(n)}\}$

$$\mathbf{r}^{(k)} = \mathbf{b} - A\mathbf{x}^{(k)}$$

Generalized minimal residuals (GMRES)

Linear system $A\mathbf{x} = \mathbf{b}$

Iterative method that minimize the residual in Krylov subspaces

$$\{\mathbf{b}, A\mathbf{b}, A^2\mathbf{b}, \dots, A^{(k-1)}\mathbf{b}\}$$

Large linear systems with nonsymmetric matrices are commonly solved using the GMRES method.